

L4 – Web database server

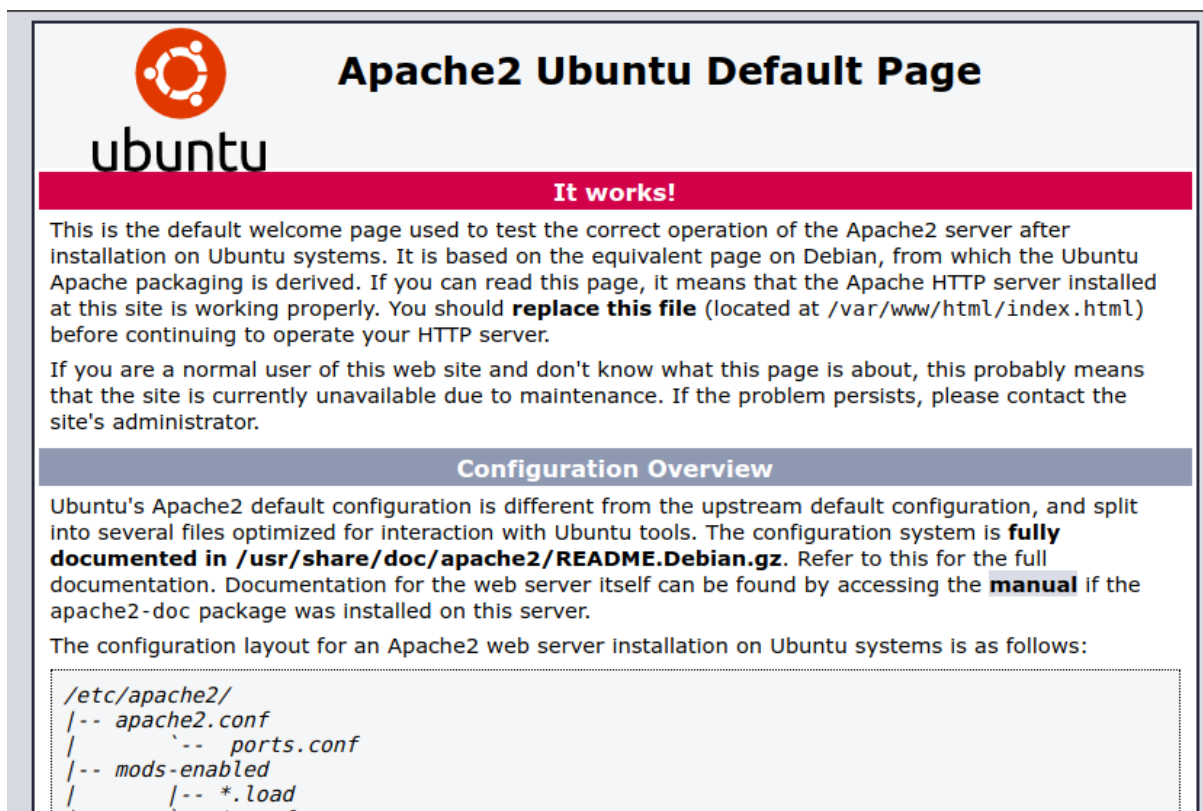
Installing Apache

To install Apache, install the latest meta-package apache2 by running:

```
sudo apt update
```

```
sudo apt install apache2
```

After letting the command run, all required packages are installed and we can test it out by typing in our IP address for the web server.



The screenshot shows the default Apache2 welcome page on Ubuntu. It features the Ubuntu logo and the text 'Apache2 Ubuntu Default Page'. A red banner with the text 'It works!' is prominently displayed. Below this, a paragraph explains that this is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It mentions that the page is based on the equivalent page on Debian and that if the user can read this page, it means the Apache HTTP server is working properly. It also advises replacing the file located at /var/www/html/index.html before continuing to operate the HTTP server. A section titled 'Configuration Overview' follows, explaining that Ubuntu's Apache2 default configuration is different from the upstream default configuration and is split into several files optimized for interaction with Ubuntu tools. It states that the configuration system is fully documented in /usr/share/doc/apache2/README.Debian.gz and refers to this for the full documentation. It also mentions that documentation for the web server itself can be found by accessing the manual if the apache2-doc package was installed on this server. Finally, it states that the configuration layout for an Apache2 web server installation on Ubuntu systems is as follows: /etc/apache2/, followed by a list of files: apache2.conf, ports.conf, mods-enabled, and *.load.

Apache2 Ubuntu Default Page

It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at /var/www/html/index.html) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

Configuration Overview

Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is **fully documented in /usr/share/doc/apache2/README.Debian.gz**. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the apache2-doc package was installed on this server.

The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:

```
/etc/apache2/  
|-- apache2.conf  
|   |-- ports.conf  
|-- mods-enabled  
|   |-- *.load
```

If you see the page above, it means that Apache has been successfully installed on your

Installing MariaDB Server Guide

Update Package List:

Before installing, it's a good practice to update your package index.

For Debian/Ubuntu:Bash

```
sudo apt update
```

Install MariaDB Server:

Install the MariaDB server and client packages.

For Debian/Ubuntu:Bash

```
sudo apt install mariadb-server mariadb-client galera-4
```

Start and Verify the Service:

MariaDB typically starts automatically after installation. You can check its status and manually start it if needed.

Check status:

```
sudo systemctl status mariadb
```

Start service:

```
sudo systemctl start mariadb
```

Verify installation by connecting as root:

```
mariadb -u root -p
```

Enter the root password you set during the secure installation.

Installing php

To install php you only need to run one command:

For the default PHP version:

```
sudo apt install php -y
```

Installing phpmyadmin

1. Download the latest version

```
wget https://www.phpmyadmin.net/downloads/phpMyAdmin-latest-all-languages.tar.gz
```

2. Extract and move it to your web folder

```
tar xvf phpMyAdmin-latest-all-languages.tar.gz
```

```
sudo mv phpMyAdmin-*-all-languages /var/www/html/phpmyadmin
```

3. Create a temporary folder for phpMyAdmin features

```
sudo mkdir -p /var/www/html/phpmyadmin/tmp
```

```
sudo chown -R www-data:www-data /var/www/html/phpmyadmin
```